ASIF SHAHRIAR

Graduate, CSE, BUET | Lecturer, BRAC University | Research Intern (Remote), PurSec Lab, Purdue

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RESEARCH INTERESTS

- Computer Security, AI Security, Vulnerability Detection, Adversarial ML
- Behavioural Security in LLMs and LLM-Agents

EDUCATION

Bangladesh University of Engineering & Technology (BUET) **2019 – 2024**

B.Sc in Computer Science & Engineering

- CGPA: **3.85/4.0** [Scholaro US-equiv: **3.91/4.0**, CSE major-only CGPA: **3.91/4.0**]
- **Thesis:** 5G Vulnerability Detection using LLMs through Zero-Shot and In-Context Learning (published in **IEEE TIFS**). Supervisor: **Prof. Md. Shohrab Hossain**

PEER-REVIEWED PUBLICATIONS

1. **[IEEE S&P '27] Asif Shahriar**, Hongyu Cai, Hadjer Benkraouda, Gang Wang, Z. Berkay Celik. "Words Speak Louder Than Code: Investigating Cognitive Heuristics in LLM-Based Code Vulnerability Detection." To appear in *Proceedings of the 48th **IEEE Symposium on Security and Privacy (IEEE S&P)***, 2027. [Paper] [GitHub]
2. **[IEEE TIFS '25] Asif Shahriar**, Syed Jarullah Hisham, K. M. Asifur Rahman, Ruhan Islam, Md. Shohrab Hossain, Ren-Hung Hwang, Ying-Dar Lin. "5GPT: 5G Vulnerability Detection by Combining Zero-Shot Capabilities of GPT-4 With Domain-Aware Strategies Through Prompt Engineering." ***IEEE Transactions on Information Forensics and Security (IEEE TIFS)***, 2025. [Paper]
3. **[EMNLP '25] Asif Shahriar**, Rifat Shahriyar, M Saifur Rahman. "Inceptive Transformers: Enhancing Contextual Representations through Multi-Scale Feature Learning Across Domains and Languages." In *Proceedings of the 2025 Conference on **Empirical Methods in Natural Language Processing (EMNLP)***, 2025. [Paper] [GitHub]
4. **[TMLR '26] Asif Shahriar**, Md Nafiu Rahman, Sadif Ahmed, Farig Sadeque, Md Rizwan Parvez. "A Survey on Agentic Security: Applications, Threats and Defenses." ***Transactions on Machine Learning Research (TMLR)***, 2026. [Paper] [GitHub]
5. **[NSysS '24] Asif Shahriar**, Debojit Pandit, M Saifur Rahman. "XLNet-CNN: Combining Global Context Understanding of XLNet with Local Context Capture through Convolution for Improved Multi-Label Text Classification." In *Proceedings of the **11th International Conference on Networking, Systems and Security (NSysS)***, 2024. Association for Computing Machinery. **Best Paper Award**. [Paper] [GitHub]

PRE-PRINTS UNDER REVIEW

1. **Asif Shahriar**, Md Nafiu Rahman, Sadif Ahmed, Farig Sadeque, Md Rizwan Parvez. "AgentTell: Behavioural Side-Channel Leakage in Browser-Use Agents." Under review at **ICLR 2027**. [Paper] [GitHub]

RESEARCH EXPERIENCE

Research Intern (Remote), PurSec Lab, Purdue University **Dec 2025 – Present**

Advisor: **Prof. Z. Berkay Celik**

- Led a study showing that LLM-based vulnerability detection is susceptible to three classic cognitive biases: halo effect, framing effect and anchoring effect
- Showed that models change their vulnerability verdict on the same code based on non-code context, such as who wrote it, task/consequence framing, and prior scanner results. Attackers can exploit these biases to suppress detection of vulnerable code in AI-assisted review tools by up to 97%
- Resulted in a first-author paper at **IEEE S&P 2027**

Research Collaboration (Remote), QCRI **Aug 2025 – Present**

Advisor: **Dr. Md Rizwan Parvez** (Scientist, QCRI)

- Led a study on defining and investigating behavioural side-channel leakage in browser-use agents, where an agent's actions on a website reveal private information from its context, without any adversarial injection or instruction
- Developed AgentTell, a benchmark of 100 tasks across 20 scenarios, and found leakage in 56–70% of tasks across six backbone models [Under review, **ICLR 2027**]
- Also led a survey on agentic security covering applications, threats, and defenses, which has been accepted in **TMLR**

TEACHING EXPERIENCE

Full-time Lecturer, Department of CSE, BRAC University **July 2024 – Present**

Course Instructor

- Artificial Intelligence (Theory and Sessional)
- Data Communications (Theory)
- Data Structures and Algorithms (Sessional)

Thesis Supervisor: Co-advising two undergraduate thesis groups on agentic security

ACADEMIC SERVICE

- **Reviewer**, EMNLP 2026
- **Reviewer**, IEEE Transactions on Cybernetics
- **Reviewer**, Cybersecurity (Springer)
- **External Reviewer**, AIDC Workshop 2026

TECHNICAL SKILLS

Languages: Python • Java • Javascript • C • C++ • LaTeX • Assembly
LLMs & Agents: RAG • fine-tuning • in-context learning • prompt engineering • tool-using agents • browser-use agents • coding agents • agentic security evaluation
Deep-Learning: PyTorch • TensorFlow • Neural Networks (ResNet, InceptionNet, ViT, RNN, LSTM) • Transformers • Language models
Systems & Tooling: Linux • Docker • Git • OracleDB • PostgreSQL
Networking & Security: 5G • LTE • Cryptography • EDR • NS2 • Open5GS • UERANSIM • 5GReplay • Scapy • CPPCheck • WireShark

HONOURS, AWARDS & GRANTS

- **Best paper award** in 11th International Conference on Networking, Systems, and Security (NSysS 2024)
- **Dean's List award** for academic excellence in undergrad
- ACL travel grant to present my paper at the main conference of **EMNLP 2025**
- **Talent-pool scholarship** in HSC from Ministry of Education, Bangladesh (Merit Position: **10th**)
- **Talent-pool scholarship** in SSC from Ministry of Education, Bangladesh (Merit Position: **26th**)

REFERENCES

- **Dr. Md. Shohrab Hossain**, *undergrad thesis supervisor, co-author*
  Professor, CSE, BUET | email: mshohrabhossain@cse.buet.ac.bd
- **Dr. Z. Berkay Celik**, *research supervisor, co-author*
  Associate Professor, Department of Computer Science, Purdue University
  email: zcelik@purdue.edu
- **Dr. Gang Wang**, *research supervisor, co-author*
  Associate Professor, Siebel School of Computing and Data Science, University of Illinois Urbana-Champaign (UIUC) | email: gangw@illinois.edu